

17 Functions of a Complex Variable

EXERCISES 17.1

Complex Numbers

1. $3 + 3i$
2. $-4i$
3. $i^8 = (i^2)^4 = (-1)^4 = 1$
4. $i^{11} = i(i^2)^5 = i(-1)^5 = -i$
5. $7 - 13i$
6. $-3 - 9i$
7. $-7 + 5i$
8. $-7 + 8i$
9. $11 - 10i$
10. $\frac{3}{4} + \frac{2}{3}i$
11. $-5 + 12i$
12. $-2 - 2i$
13. $-2i$
14. $\frac{i}{1+i} \cdot \frac{1-i}{1-i} = \frac{i+1}{2} = \frac{1}{2} + \frac{1}{2}i$
15. $\frac{2-4i}{3+5i} \cdot \frac{3-5i}{3-5i} = \frac{-14-22i}{34} = -\frac{7}{17} - \frac{11}{17}i$
16. $\frac{10-5i}{6+2i} \cdot \frac{6-2i}{6-2i} = \frac{50-50i}{40} = \frac{5}{4} - \frac{5}{4}i$
17. $\frac{9+7i}{1+i} \cdot \frac{1-i}{1-i} = \frac{16-2i}{2} = 8 - i$
18. $\frac{3-i}{11-2i} \cdot \frac{11+2i}{11+2i} = \frac{35-5i}{125} = \frac{7}{25} - \frac{1}{25}i$
19. $\frac{2-11i}{6-i} \cdot \frac{6+i}{6+i} = \frac{23-64i}{37} = \frac{23}{37} - \frac{64}{37}i$
20. $\frac{4+3i}{3+4i} \cdot \frac{3-4i}{3-4i} = \frac{24-7i}{25} = \frac{24}{25} - \frac{7}{25}i$
21. $(1+i)(10+10i) = 10(1+i)^2 = 20i$
22. $[(1+i)(1-i)]^2(1-i) = 4 - 4i$
23. $20 + 23i + \frac{1}{2-i} \cdot \frac{2+i}{2+i} = 20 + 23i + \frac{2}{5} + \frac{1}{5}i = \frac{102}{5} + \frac{116}{5}i$
24. $(2+3i)(-i)^2 = -2 - 3i$
25. $\frac{i}{9+7i} \cdot \frac{9-7i}{9-7i} = \frac{7+9i}{130} = \frac{7}{130} + \frac{9}{130}i$
26. $\frac{1}{6+8i} \cdot \frac{6-8i}{6-8i} = \frac{6-8i}{84} = \frac{1}{14} - \frac{2}{21}i$

27. $\frac{x}{x^2 + y^2}$ 28. $x^2 - y^2$
29. $-2y - 4$ 30. 0
31. $\sqrt{(x-1)^2 + (y-3)^2}$ 32. $\sqrt{36x^2 + 16y^2}$
33. $2x + 2yi = -9 + 2i$ implies $2x = -9$ and $2y = 2$. Hence $z = -\frac{9}{2} + i$.
34. $-x + 3yi = -7 + 6i$ implies $-x = -7$ and $3y = 6$. Hence $z = 7 + 2i$.
35. $x^2 - y^2 + 2xyi = 0 + i$ implies $x^2 - y^2 = 0$ and $2xy = 1$. Now $y = x$ implies $2x^2 = 1$ and so $x = \pm 1/\sqrt{2}$. The choice $y = -x$ gives $-2x^2 = 1$ which has no real solution. Hence $z = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i$ and $z = -\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}i$.
36. $x^2 - y^2 - 4x + (-2xy - 4y)i = 0 + 0i$ implies $x^2 - y^2 - 4x = 0$ and $y(-2x - 4) = 0$. If $y = 0$ then $x(x - 4) = 0$ and so $z = 0$ and $z = 4$. If $-2x - 4 = 0$ or $x = -2$ then $12 - y^2 = 0$ or $y = \pm 2\sqrt{3}$. This gives $z = -2 + 2\sqrt{3}i$ and $z = -2 - 2\sqrt{3}i$.
37. $|10 + 8i| = \sqrt{164}$ and $|11 - 6i| = \sqrt{157}$. Hence $11 - 6i$ is closer to the origin.
38. $|\frac{1}{2} - \frac{1}{4}i| = \frac{\sqrt{5}}{4}$ and $|\frac{2}{3} + \frac{1}{6}i| = \frac{\sqrt{17}}{6}$. Since $\frac{\sqrt{5}}{4} < \frac{\sqrt{17}}{6}$, $\frac{1}{2} - \frac{1}{4}i$ is closer to the origin.
39. $|z_1 - z_2| = |(x_1 - x_2) + i(y_1 - y_2)| = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$ which is the distance formula in the plane.
40. By the triangle inequality, $|z + 6 + 8i| \leq |z| + |6 + 8i|$. On the circle, $|z| = 2$ and so $|z + 6 + 8i| \leq 2 + \sqrt{100} = 12$.

EXERCISES 17.2

Powers and Roots

1. $2(\cos 2\pi + i \sin 2\pi)$ 2. $10(\cos \pi + i \sin \pi)$
3. $3\left(\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2}\right)$ 4. $6\left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}\right)$
5. $\sqrt{2}\left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4}\right)$ 6. $5\sqrt{2}\left(\cos \frac{7\pi}{4} + i \sin \frac{7\pi}{4}\right)$
7. $2\left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6}\right)$ 8. $4\left(\cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3}\right)$
9. $\frac{3\sqrt{2}}{2}\left(\cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4}\right)$ 10. $6\left[\cos\left(-\frac{\pi}{6}\right) + i \sin\left(-\frac{\pi}{6}\right)\right]$
11. $z = -\frac{5\sqrt{3}}{2} - \frac{5}{2}i$ 12. $z = -8 + 8i$
13. $z = 5.5433 + 2.2961i$ 14. $z = 8.0902 + 5.8779i$
15. $z_1 z_2 = 8\left[\cos\left(\frac{\pi}{8} + \frac{3\pi}{8}\right) + i \sin\left(\frac{\pi}{8} + \frac{3\pi}{8}\right)\right] = 8i$; $\frac{z_1}{z_2} = \frac{1}{2}\left[\cos\left(\frac{\pi}{8} - \frac{3\pi}{8}\right) + i \sin\left(\frac{\pi}{8} - \frac{3\pi}{8}\right)\right] = \frac{\sqrt{2}}{4} - \frac{\sqrt{2}}{4}i$

17.2 Powers and Roots

$$16. z_1 z_2 = \sqrt{6} \left[\cos \left(\frac{\pi}{4} + \frac{\pi}{12} \right) + i \sin \left(\frac{\pi}{4} + \frac{\pi}{12} \right) \right] = \frac{\sqrt{6}}{2} + \frac{3\sqrt{2}}{2} i$$

$$\frac{z_1}{z_2} = \frac{\sqrt{6}}{3} \left[\cos \left(\frac{\pi}{4} - \frac{\pi}{12} \right) + i \sin \left(\frac{\pi}{4} - \frac{\pi}{12} \right) \right] = \frac{\sqrt{2}}{2} + \frac{\sqrt{6}}{6} i$$

$$17. \left[3\sqrt{2} \left(\cos \frac{7\pi}{4} + i \sin \frac{7\pi}{4} \right) \right] \left[10 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right) \right] = 30\sqrt{2} \left[\cos \left(\frac{7\pi}{4} + \frac{\pi}{3} \right) + i \sin \left(\frac{7\pi}{4} + \frac{\pi}{3} \right) \right]$$

$$= 40.9808 + 10.9808i$$

$$18. \left[4\sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right) \right] \left[\sqrt{2} \left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} \right) \right] = 8 \left[\cos \left(\frac{\pi}{4} + \frac{3\pi}{4} \right) + i \sin \left(\frac{\pi}{4} + \frac{3\pi}{4} \right) \right] = -8$$

$$19. \frac{\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2}}{2\sqrt{2} \left[\cos \frac{7\pi}{4} + i \sin \frac{7\pi}{4} \right]} = \frac{\sqrt{2}}{4} \left[\cos \left(\frac{3\pi}{2} - \frac{7\pi}{4} \right) + i \sin \left(\frac{3\pi}{2} - \frac{7\pi}{4} \right) \right] = \frac{1}{4} - \frac{1}{4} i$$

$$20. \frac{2\sqrt{2} \left[\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right]}{2 \left[\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right]} = \sqrt{2} \left[\cos \left(\frac{\pi}{3} - \frac{2\pi}{3} \right) + i \sin \left(\frac{\pi}{3} - \frac{2\pi}{3} \right) \right] = \frac{\sqrt{2}}{2} - \frac{\sqrt{6}}{2} i$$

$$21. 2^9 \left[\cos \frac{9\pi}{3} + i \sin \frac{9\pi}{3} \right] = -512$$

$$22. (2\sqrt{2})^5 \left[\cos \left(-\frac{5\pi}{4} \right) + i \sin \left(-\frac{5\pi}{4} \right) \right] = -128 + 128i$$

$$23. \left(\frac{\sqrt{2}}{2} \right)^{10} \left[\cos \frac{10\pi}{4} + i \sin \frac{10\pi}{4} \right] = \frac{1}{32} i$$

$$24. (2\sqrt{2})^4 \left[\cos \frac{8\pi}{3} + i \sin \frac{8\pi}{3} \right] = -32 + 32\sqrt{3} i$$

$$25. \cos \frac{12\pi}{8} + i \sin \frac{12\pi}{8} = -i$$

$$26. (\sqrt{3})^6 \left[\cos \frac{12\pi}{9} + i \sin \frac{12\pi}{9} \right] = -\frac{27}{2} - \frac{27\sqrt{3}}{2} i$$

$$27. 8^{1/3} = 2 \left[\cos \frac{2k\pi}{3} + i \sin \frac{2k\pi}{3} \right], \quad k = 0, 1, 2$$

$$w_0 = 2[\cos 0 + i \sin 0] = 2; \quad w_1 = 2 \left[\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right] = -1 + \sqrt{3} i$$

$$w_2 = 2 \left[\cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3} \right] = -1 - \sqrt{3} i$$

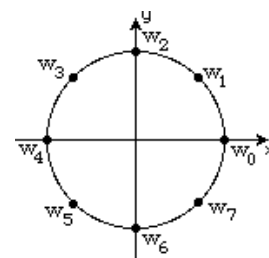
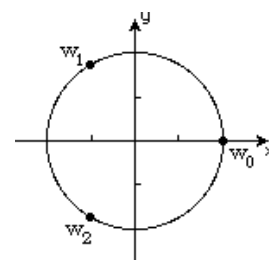
$$28. (1)^{1/8} = \cos \frac{k\pi}{4} + i \sin \frac{k\pi}{4}, \quad k = 0, 1, 2, \dots, 7$$

$$w_0 = \cos 0 + i \sin 0 = 1; \quad w_1 = \cos \frac{\pi}{4} + i \sin \frac{\pi}{4} = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} i$$

$$w_2 = \cos \frac{\pi}{2} + i \sin \frac{\pi}{2} = i; \quad w_3 = \cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} = -\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} i$$

$$w_4 = \cos \pi + i \sin \pi = -1; \quad w_5 = \cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4} = -\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} i$$

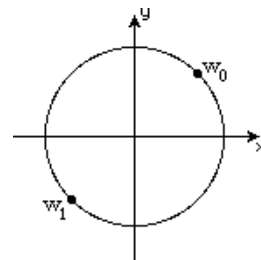
$$w_6 = \cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2} = -i; \quad w_7 = \cos \frac{7\pi}{4} + i \sin \frac{7\pi}{4} = \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} i$$



29. $(i)^{1/2} = \cos\left(\frac{\pi}{4} + k\pi\right) + i \sin\left(\frac{\pi}{4} + k\pi\right), \quad k = 0, 1$

$$w_0 = \cos \frac{\pi}{4} + i \sin \frac{\pi}{4} = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} i$$

$$w_1 = \cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4} = -\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} i$$

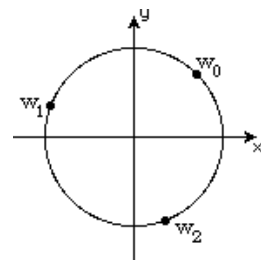


30. $(-1 + i)^{1/3} = 2^{1/6} \left[\cos\left(\frac{\pi}{4} + \frac{2k\pi}{3}\right) + i \sin\left(\frac{\pi}{4} + \frac{2k\pi}{3}\right) \right], \quad k = 0, 1, 2$

$$w_0 = 2^{1/6} \left[\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right] = \frac{1}{\sqrt[3]{2}} + \frac{1}{\sqrt[3]{2}} i = 0.7937 + 0.7937i$$

$$w_1 = 2^{1/6} \left[\cos \frac{11\pi}{12} + i \sin \frac{11\pi}{12} \right] = -1.0842 + 0.2905i$$

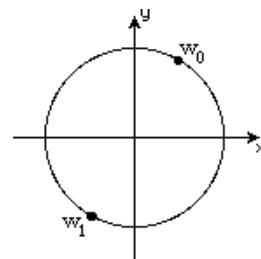
$$w_2 = 2^{1/6} \left[\cos \frac{19\pi}{12} + i \sin \frac{19\pi}{12} \right] = 0.2905 - 1.0842i$$



31. $(-1 + \sqrt{3}i)^{1/2} = 2^{1/2} \left[\cos\left(\frac{\pi}{3} + k\pi\right) + i \sin\left(\frac{\pi}{3} + k\pi\right) \right], \quad k = 0, 1$

$$w_0 = 2^{1/2} \left[\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right] = \frac{\sqrt{2}}{2} + \frac{\sqrt{6}}{2} i$$

$$w_2 = 2^{1/2} \left[\cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3} \right] = -\frac{\sqrt{2}}{2} - \frac{\sqrt{6}}{2} i$$

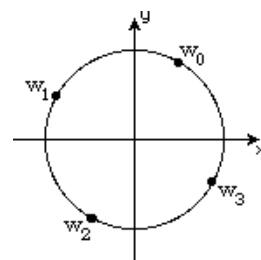


32. $(-1 - \sqrt{3}i)^{1/4} = 2^{1/4} \left[\cos\left(\frac{\pi}{3} + \frac{k\pi}{2}\right) + i \sin\left(\frac{\pi}{3} + \frac{k\pi}{2}\right) \right], \quad k = 0, 1, 2, 3$

$$w_0 = 2^{1/4} \left[\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right] = 2^{1/4} \left[\frac{1}{2} + \frac{\sqrt{3}}{2} i \right]$$

$$w_1 = 2^{1/4} \left[\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} \right] = 2^{1/4} \left[-\frac{\sqrt{3}}{2} + \frac{1}{2} i \right]$$

$$w_2 = 2^{1/4} \left[\cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3} \right] = 2^{1/4} \left[-\frac{1}{2} - \frac{\sqrt{3}}{2} i \right]; \quad w_3 = 2^{1/4} \left[\cos \frac{11\pi}{6} + i \sin \frac{11\pi}{6} \right] = 2^{1/4} \left[\frac{\sqrt{3}}{2} - \frac{1}{2} i \right]$$



33. The solutions are the four fourth roots of -1 ;

$$w_k = \cos \frac{\pi + 2k\pi}{4} + i \sin \frac{\pi + 2k\pi}{4}, \quad k = 0, 1, 2, 3.$$

We have

$$w_1 = \cos \frac{\pi}{4} + i \sin \frac{\pi}{4} = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} i$$

$$w_3 = \cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4} = -\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} i$$

$$w_2 = \cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} = -\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} i$$

$$w_4 = \cos \frac{7\pi}{4} + i \sin \frac{7\pi}{4} = \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} i.$$

34. $(z^4 - 1)^2 = 0$ is the same as $(z - i)^2(z + i)^2(z - 1)^2(z + 1)^2 = 0$. Thus $z_1 = 1$, $z_2 = -1$, $z_3 = i$, and $z_4 = -i$ are roots of multiplicity two.

17.2 Powers and Roots

$$\begin{aligned}
 35. \quad \left(\cos \frac{\pi}{9} + i \sin \frac{\pi}{9}\right)^{12} \left[2 \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right)\right]^5 &= 2^5 \left(\cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3}\right) \left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6}\right) \\
 &= 32 \left[\cos \left(\frac{4\pi}{3} + \frac{5\pi}{6}\right) + i \sin \left(\frac{4\pi}{3} + \frac{5\pi}{6}\right)\right] \\
 &= 32 \left(\cos \frac{13\pi}{6} + i \sin \frac{13\pi}{6}\right) = 32 \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right) = 16\sqrt{3} + 16i
 \end{aligned}$$

$$36. \quad \frac{\left[8 \left(\cos \frac{3\pi}{8} + i \sin \frac{3\pi}{8}\right)\right]^3}{\left[2 \left(\cos \frac{\pi}{16} + i \sin \frac{\pi}{16}\right)\right]^{10}} = \frac{2^9}{2^{10}} \left[\cos \left(\frac{9\pi}{8} - \frac{10\pi}{16}\right) + i \left(\frac{9\pi}{8} - \frac{10\pi}{16}\right)\right] = \frac{1}{2} \left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}\right) = \frac{1}{2} i$$

37. We have

$$(\cos 2\theta + i \sin \theta)^2 = \cos 2\theta + i \sin 2\theta$$

Also

$$(\cos \theta + i \sin \theta)^2 = \cos^2 \theta - \sin^2 \theta + (2 \sin \theta \cos \theta)i.$$

Equating real and imaginary parts gives

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta, \quad \sin 2\theta = 2 \sin \theta \cos \theta.$$

38. We have

$$(\cos \theta + i \sin \theta)^3 = \cos 3\theta + i \sin 3\theta.$$

Also

$$\begin{aligned}
 (\cos \theta + i \sin \theta)^3 &= \cos^3 \theta + 3 \cos^2 \theta (i \sin \theta) + 3 \cos \theta (i \sin \theta)^2 + (i \sin \theta)^3 \\
 &= \cos^3 \theta - 3 \cos \theta \sin^2 \theta + (3 \cos^2 \theta \sin \theta - \sin^3 \theta)i.
 \end{aligned}$$

Equating real and imaginary parts gives

$$\cos 3\theta = \cos^3 \theta - 3 \cos \theta \sin^2 \theta, \quad \sin 3\theta = 3 \cos^2 \theta \sin \theta - \sin^3 \theta.$$

$$39. \quad (a) \quad \text{Arg}(z_1) = \pi, \quad \text{Arg}(z_2) = \frac{\pi}{2}, \quad \text{Arg}(z_1 z_2) = -\frac{\pi}{2}, \quad \text{Arg}(z_1) + \text{Arg}(z_2) = \frac{3\pi}{2} \neq \text{Arg}(z_1 z_2)$$

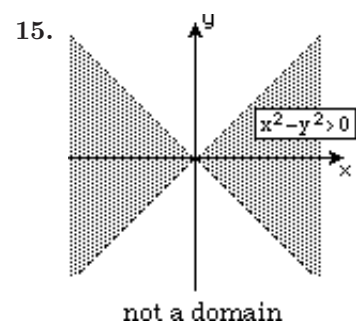
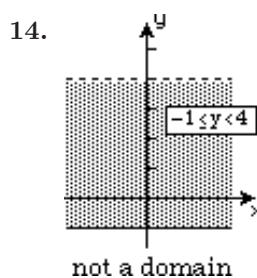
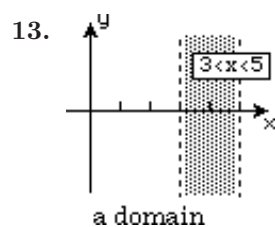
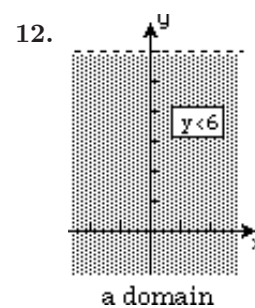
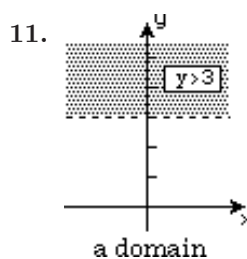
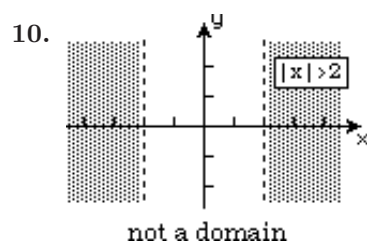
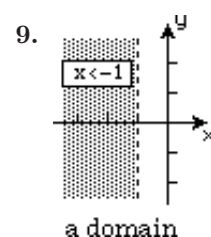
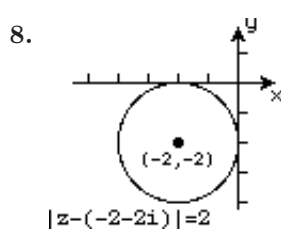
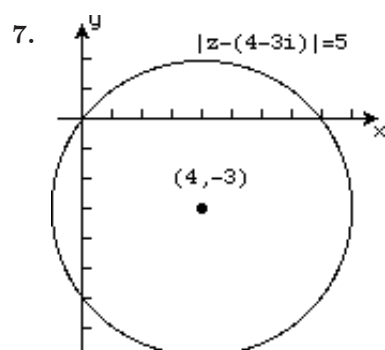
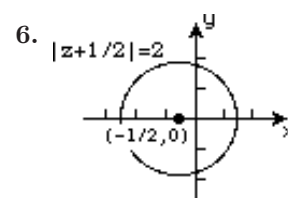
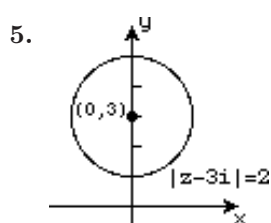
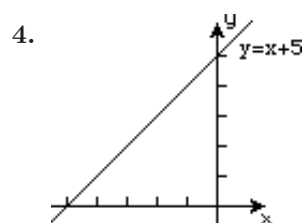
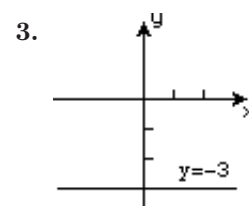
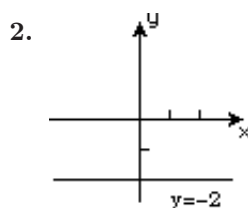
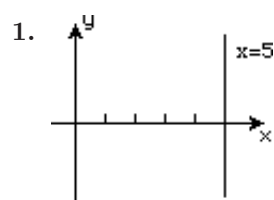
$$(b) \quad \text{Arg}(z_1/z_2) = -\frac{\pi}{2}, \quad \text{Arg}(z_1) - \text{Arg}(z_2) = \pi - \frac{\pi}{2} = \frac{\pi}{2} \neq \text{Arg}(z_1/z_2)$$

40. (a) If we take $\arg(z_1) = \pi$ and $\arg(z_2) = \pi/2$ then $\arg(z_1) + \arg(z_2) = 3\pi/2$ is an argument of the product $z_1 z_2 = -5i$. With these same arguments we see that $\arg(z_1) - \arg(z_2) = \pi/2$ is an argument of the quotient $z_1/z_2 = \frac{1}{5}i$.

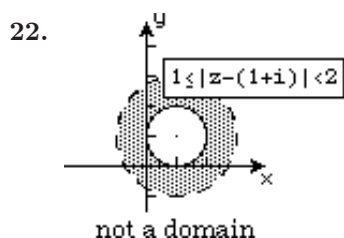
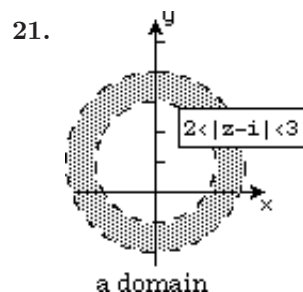
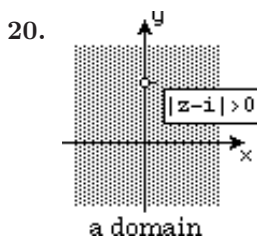
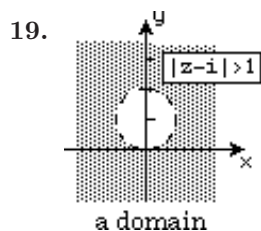
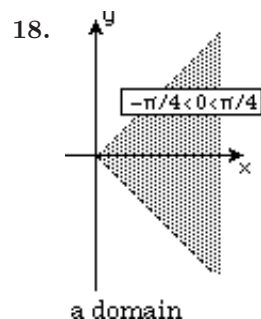
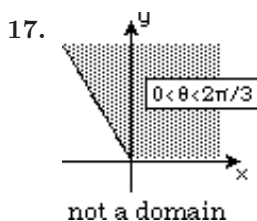
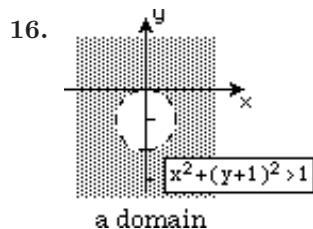
(b) If we take $\arg(z_1) = \pi$ and $\arg(z_2) = -\pi/2$ then $\arg(z_1) + \arg(z_2) = \pi/2$ is an argument of the product $z_1 z_2 = 5i$. With these same arguments we see that $\arg(z_1) - \arg(z_2) = 3\pi/2$ is an argument of the quotient $z_1/z_2 = -\frac{1}{5}i$.

EXERCISES 17.3

Sets in the Complex Plane



17.3 Sets in the Complex Plane

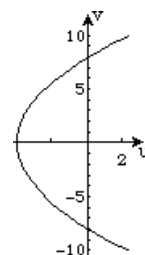


23. The given equation is equivalent to $(x+1)^2 + y^2 = x^2 + (y-1)^2$. This simplifies to $y = -x$ which describes a straight line through the origin.
24. $|\operatorname{Re}(z)| = |x|$ is the same as $\sqrt{x^2}$ and $|z| = \sqrt{x^2 + y^2}$. Since $y^2 \geq 0$ the inequality $\sqrt{x^2} \leq \sqrt{x^2 + y^2}$ is true for all complex numbers.
25. The given equation simplifies to the equation $x^2 - y^2 = 1$ which is a hyperbola with center at the origin.
26. Since $|z-i|$ and $|z-(-i)|$ represent distances from the point (x, y) to i and $-i$, respectively, the equation is the distance formula definition of an ellipse with foci at $(0, 1)$ and $(0, -1)$.

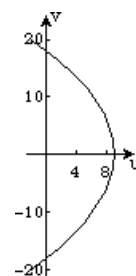
EXERCISES 17.4

Functions of a Complex Variable

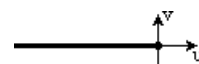
1. Substituting $y = 2$ into $u = x^2 - y^2$, $v = 2xy$ gives the parametric equations $u = x^2 - 4$, $v = 4x$. Using $x = v/4$ the first equation gives $u = v^2/16 - 4$. The graph is the parabola shown.



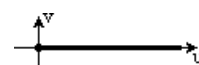
2. Substituting $x = -3$ into $u = x^2 - y^2$, $v = 2xy$ gives the parametric equations $u = 9 - y^2$, $v = -6y$. Using $y = -v/6$ the first equation gives $u = 9 - v^2/36$. The graph is the parabola shown.



3. $x = 0$ gives $u = -y^2$, $v = 0$. Since $-y^2 \leq 0$ for all real values of y , the image is the origin and the negative u -axis.



4. $y = 0$ gives $u = x^2$, $v = 0$. Since $x^2 \geq 0$ for all real values of x , the image is the origin and the positive u -axis.



5. $y = x$ gives $u = 0$, $v = 2x^2$. Since $x^2 \geq 0$ for all real values of x , the image is the origin and the positive v -axis.



6. $y = -x$ gives $u = 0$, $v = -2x^2$. Since $-x^2 \leq 0$ for all real values of x , the image is the origin and the negative v -axis.



7. $f(z) = (6x - 5) + i(6y + 9)$

8. $f(z) = (7x - 9y - 3) + i(7y - 9x + 2)$

9. $f(z) = (x^2 - y^2 - 3x) + i(2xy - 3y + 4)$

10. $f(z) = (3x^2 - 3y^2 + 2x) + i(-6xy + 2y)$

11. $f(z) = (x^3 - 3xy^2 - 4x) + i(3x^2y - y^3 - 4y)$

12. $f(z) = (x^4 - 6x^2y^2 + y^4) + i(4x^3y - 4xy^3)$

17.4 Functions of a Complex Variable

13. $f(z) = \left(x + \frac{x}{x^2 + y^2}\right) + i\left(y - \frac{y}{x^2 + y^2}\right)$ 14. $f(z) = \frac{x^2 + y^2 + x}{(x+1)^2 + y^2} + i\frac{y}{(x+1)^2 + y^2}$
15. (a) $f(0 + 2i) = -4 + i$ (b) $f(2 - i) = 3 - 9i$ (c) $f(5 + 3i) = 1 + 86i$
16. (a) $f(1 + i) = 3 - 2i$ (b) $f(2 - i) = \frac{7}{2} + 10i$ (c) $f(1 + 4i) = 3 - 32i$
17. (a) $f(4 - 6i) = 14 - 20i$ (b) $f(-5 + 12i) = -13 + 43i$ (c) $f(2 - 7i) = 3 - 26i$
18. (a) $f(0 + \frac{\pi}{4}i) = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i$ (b) $f(-1 - \pi i) = -e^{-1}$ (c) $f(3 + \frac{\pi}{3}i) = \frac{1}{2}e^3 + \frac{\sqrt{3}}{2}e^3i$
19. $\lim_{z \rightarrow i}(4z^3 - 5z^2 + 4z + 1 - 5i) = 6 - 5i$
20. $\lim_{z \rightarrow 1-i} \frac{5z^2 - 2z + 2}{z + 1} = \frac{5(1-i)^2 - 2(1-i) + 2}{2-i} = \frac{8}{5} - \frac{16}{5}i$
21. $\lim_{z \rightarrow i} \frac{z^4 - 1}{z - i} = \lim_{z \rightarrow i} \frac{(z^2 - 1)(z - i)(z + i)}{z - i} = -4i$
22. $\lim_{z \rightarrow 1+i} \frac{z^2 - 2z + 2}{z^2 - 2i} = \lim_{z \rightarrow 1+i} \frac{[z - (1+i)][z - (1-i)]}{[z - (1+i)][z - (-1-i)]} = \frac{1}{2} + \frac{1}{2}i$
23. Along the y -axis, $\lim_{z \rightarrow 0} \frac{x + iy}{x - iy} = \lim_{y \rightarrow 0} \frac{iy}{-iy} = -1$, whereas along the x -axis, $\lim_{z \rightarrow 0} \frac{x + iy}{x - iy} = \lim_{x \rightarrow 0} \frac{x}{x} = 1$.
24. Along the line $x = 1$, $\lim_{z \rightarrow 1} \frac{x + y - 1}{z - 1} = \lim_{y \rightarrow 0} \frac{y}{iy} = \frac{1}{i} = -i$, whereas along the x -axis, $\lim_{z \rightarrow 1} \frac{x + y - 1}{z - 1} = \lim_{x \rightarrow 1} \frac{x - 1}{x - 1} = 1$.
25. $f'(z) = \lim_{\Delta z \rightarrow 0} \frac{(z + \Delta z)^2 - z^2}{\Delta z} = \lim_{\Delta z \rightarrow 0} \frac{2z\Delta z + (\Delta z)^2}{\Delta z} = \lim_{\Delta z \rightarrow 0} (2z + \Delta z) = 2z$
26. $f'(z) = \lim_{\Delta z \rightarrow 0} \frac{\frac{1}{z + \Delta z} - \frac{1}{z}}{\Delta z} = \lim_{\Delta z \rightarrow 0} \frac{-\Delta z}{(\Delta z)z(z + \Delta z)} = \lim_{\Delta z \rightarrow 0} \frac{-1}{z(z + \Delta z)} = -\frac{1}{z^2}$
27. $f'(z) = 12z^2 - (6 + 2i)z - 5$
28. $f'(z) = 20z^3 - 3iz^2 + (16 - 2i)z$
29. $f'(z) = (2z + 1)(2z - 4) + 2(z^2 - 4z + 8i) = 6z^2 - 14z - 4 + 8i$
30. $f'(z) = (z^5 + 3iz^3)(4z^2 + 3iz^2 + 4z - 6i) + (z^4 + iz^3 + 2z^2 - 6iz)(5z^4 + 9iz^2)$
31. $f'(z) = 6z(z^2 - 4i)^2$
32. $f'(z) = 6(2z - 1/z)^5(2 + 1/z^2)$
33. $f'(z) = \frac{(2z + i)3 - (3z - 4 + 8i)2}{(2z + 1)^2} = \frac{8 - 13i}{(2z + i)^2}$
34. $f'(z) = \frac{(z^3 + 1)(10z - 1) - (5z^2 - z)3z^2}{(z^3 + 1)^2} = \frac{-5z^4 + 2z^3 + 10z - 1}{(z^3 + 1)^2}$
35. $3i$ 36. $0, 2 - 5i$ 37. $-2i, 2i$ 38. $3 - 4i, 3 + 4i$
39. We have $\lim_{\Delta z \rightarrow 0} \frac{\overline{z + \Delta z} - \bar{z}}{\Delta z} = \lim_{\Delta z \rightarrow 0} \frac{\overline{\Delta z}}{\Delta z}$.

If we let $\Delta z \rightarrow 0$ along a horizontal line then $\Delta z = \Delta x$, $\overline{\Delta z} = \Delta x$, and

$$\lim_{\Delta z \rightarrow 0} \frac{\overline{\Delta z}}{\Delta z} = \lim_{\Delta x \rightarrow 0} \frac{\Delta x}{\Delta x} = 1.$$

If we let $\Delta z \rightarrow 0$ along a vertical line then $\Delta z = i\Delta y$, $\overline{\Delta z} = -i\Delta y$, and

$$\lim_{\Delta z \rightarrow 0} \frac{\overline{\Delta z}}{\Delta z} = \lim_{\Delta y \rightarrow 0} \frac{-i\Delta y}{i\Delta y} = -1.$$

Since these two limits are not equal, $f(z) = \bar{z}$ cannot be differentiable at any z .

40. We have
$$f'(z) = \lim_{\Delta z \rightarrow 0} \frac{(z + \Delta z)(\bar{z} + \overline{\Delta z}) - z\bar{z}}{\Delta z} = \lim_{\Delta z \rightarrow 0} \left(\bar{z} + z \frac{\overline{\Delta z}}{\Delta z} + \overline{\Delta z} \right).$$

If $z = 0$, then the above limit becomes

$$f'(0) = \lim_{\Delta z \rightarrow 0} \overline{\Delta z} = 0.$$

If $z \neq 0$ then we first let $\Delta z \rightarrow 0$ along a horizontal line so that $\Delta z = \Delta x$ and $\overline{\Delta z} = \Delta x$. Thus,

$$f'(z) = \lim_{\Delta z \rightarrow 0} \left(\bar{z} + z \frac{\Delta x}{\Delta x} + \Delta x \right) = \bar{z} + z.$$

Next we let $\Delta z \rightarrow 0$ along a vertical line so that $\Delta z = i\Delta y$, $\overline{\Delta z} = -i\Delta y$. Thus

$$f'(z) = \lim_{\Delta y \rightarrow 0} \left(\bar{z} + z \frac{-i\Delta y}{i\Delta y} + i\Delta y \right) = \bar{z} - z.$$

We must have $\bar{z} + z = \bar{z} - z$ which implies $z = 0$. This is a contradiction to the assumption that $z \neq 0$. Hence $f(z) = |z|^2$ is differentiable only at $z = 0$.

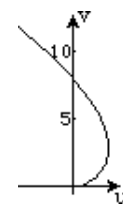
41. Each linear equation in the system $\frac{dx}{dt} = 2x$, $\frac{dy}{dt} = 2y$ can be solved directly. We obtain $x(t) = c_1 e^{2t}$ and $y(t) = c_2 e^{2t}$.

42. The system $\frac{dx}{dt} = -y$, $\frac{dy}{dt} = x$ can be solved as in Section 3.11. We obtain $x(t) = c_1 \cos t + c_2 \sin t$, $y(t) = c_1 \sin t - c_2 \cos t$.

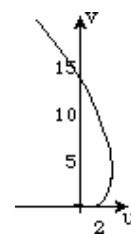
43. The equations in the system $\frac{dx}{dt} = \frac{x}{x^2 + y^2}$, $\frac{dy}{dt} = \frac{y}{x^2 + y^2}$ can be divided to give $\frac{dy}{dx} = \frac{y}{x}$. By separation of variables we obtain $y = cx$.

44. Each equation in the system $\frac{dx}{dt} = x^2$, $\frac{dy}{dt} = -y^2$ can be solved directly by separation of variables. We obtain $x(t) = \frac{-1}{t + c_1}$, $y = \frac{1}{t + c_2}$.

45. If $y = \frac{1}{2}x^2$ the equations $u = x^2 - y^2$, $v = 2xy$ give $u = x^2 - \frac{1}{4}x^4$, $v = x^3$. With the aid of a computer, the graph of these parametric equations is shown.



46. If $y = (x - 1)^2$ the equations $u = x^2 - y^2$, $v = 2xy$ give $u = x^2 - (x - 1)^4$, $v = 2x(x - 1)^2$. With the aid of a computer the graph of these parametric equations is shown.



EXERCISES 17.5

Cauchy-Riemann Equations

1. $u = x^3 - 3xy^2$, $v = 3x^2y - y^3$; $\frac{\partial u}{\partial x} = 3x^2 - 3y^2 = \frac{\partial v}{\partial y}$, $\frac{\partial u}{\partial y} = -6xy = -\frac{\partial v}{\partial x}$
2. $u = 3x^2 - 3y^2 + 5x$, $v = 6xy + 5y - 6$; $\frac{\partial u}{\partial x} = 6x + 5 = \frac{\partial v}{\partial y}$, $\frac{\partial u}{\partial y} = -6y = -\frac{\partial v}{\partial x}$
3. $u = x$, $v = 0$; $\frac{\partial u}{\partial x} = 1$, $\frac{\partial v}{\partial y} = 0$. Since $1 \neq 0$, f is not analytic at any point.
4. $u = y$, $v = x$; $\frac{\partial u}{\partial x} = 0 = \frac{\partial v}{\partial y}$, $\frac{\partial u}{\partial y} = 1$, $-\frac{\partial v}{\partial x} = -1$. Since $1 \neq -1$, f is not analytic at any point.
5. $u = -2x + 3$, $v = 10y$; $\frac{\partial u}{\partial x} = -2$, $\frac{\partial v}{\partial y} = 10$. Since $-2 \neq 10$, f is not analytic at any point.
6. $u = x^2 - y^2$, $v = -2xy$; $\frac{\partial u}{\partial x} = 2x$, $\frac{\partial v}{\partial y} = -2x$; $\frac{\partial u}{\partial y} = -2y$, $-\frac{\partial v}{\partial x} = 2y$

The Cauchy-Riemann equations hold only at $(0, 0)$. Since there is no neighborhood about $z = 0$ within which f is differentiable we conclude f is nowhere analytic.

7. $u = x^2 + y^2$, $v = 0$; $\frac{\partial u}{\partial x} = 2x$, $\frac{\partial v}{\partial y} = 0$; $\frac{\partial u}{\partial y} = 2y$, $-\frac{\partial v}{\partial x} = 0$

The Cauchy-Riemann equations hold only at $(0, 0)$. Since there is no neighborhood about $z = 0$ within which f is differentiable we conclude f is nowhere analytic.

8. $u = \frac{x}{x^2 + y^2}$, $v = \frac{y}{x^2 + y^2}$; $\frac{\partial u}{\partial x} = \frac{y^2 - x^2}{(x^2 + y^2)^2}$, $\frac{\partial v}{\partial y} = \frac{x^2 - y^2}{(x^2 + y^2)^2}$; $\frac{\partial u}{\partial y} = -\frac{2xy}{(x^2 + y^2)^2} = \frac{\partial v}{\partial x}$

The Cauchy-Riemann equations hold only at $(0, 0)$. Since there is no neighborhood about $z = 0$ within which f is differentiable, we conclude f is nowhere analytic.

9. $u = e^x \cos y$, $v = e^x \sin y$; $\frac{\partial u}{\partial x} = e^x \cos y = \frac{\partial v}{\partial y}$; $\frac{\partial u}{\partial y} = -e^x \sin y = -\frac{\partial v}{\partial x}$. f is analytic for all z .
10. $u = x + \sin x \cosh y$, $v = y + \cos x \sinh y$; $\frac{\partial u}{\partial x} = 1 + \cos x \cosh y = \frac{\partial v}{\partial y}$; $\frac{\partial u}{\partial y} = \sin x \sinh y = -\frac{\partial v}{\partial x}$.
 f is analytic for all z .

11. $u = e^{x^2 - y^2} \cos 2xy$, $v = e^{x^2 - y^2} \sin 2xy$; $\frac{\partial u}{\partial x} = -2ye^{x^2 - y^2} \sin 2xy + 2xe^{x^2 - y^2} \cos 2xy = \frac{\partial v}{\partial y}$;
 $\frac{\partial u}{\partial y} = -2xe^{x^2 - y^2} \sin 2xy - 2ye^{x^2 - y^2} \cos 2xy = -\frac{\partial v}{\partial x}$. f is analytic for all z .

12. $u = 4x^2 + 5x - 4y^2 + 9$, $v = 8xy + 5y - 1$; $\frac{\partial u}{\partial x} = 8x + 5 = \frac{\partial v}{\partial y}$, $\frac{\partial u}{\partial y} = -8y = -\frac{\partial v}{\partial x}$. f is analytic for all z .

13. $u = \frac{x-1}{(x-1)^2 + y^2}$, $v = -\frac{y}{(x-1)^2 + y^2}$; $\frac{\partial u}{\partial x} = \frac{y^2 - (x-1)^2}{[(x-1)^2 + y^2]^2} = \frac{\partial v}{\partial y}$, $\frac{\partial u}{\partial y} = -\frac{2y(x-1)}{[(x-1)^2 + y^2]^2} = -\frac{\partial v}{\partial x}$
 f is analytic in any domain not containing $z = 1$.

14. $u = \frac{x^3 + xy^2 + x}{x^2 + y^2}$, $v = \frac{x^2y + y^3 - y}{x^2 + y^2}$; $\frac{\partial u}{\partial x} = \frac{x^4 + 2x^2y^2 - x^2 + y^2 + y^4}{(x^2 + y^2)^2} = \frac{\partial v}{\partial y}$, $\frac{\partial u}{\partial y} = \frac{-2xy}{(x^2 + y^2)^2} = -\frac{\partial v}{\partial x}$
 f is analytic in any domain not containing $z = 0$.

15. $\frac{\partial u}{\partial x} = 3 = b = \frac{\partial v}{\partial y}$; $\frac{\partial u}{\partial y} = -1 = -a = -\frac{\partial v}{\partial x}$. f is analytic for all z when $b = 3$, $a = 1$.

16. The Cauchy-Riemann equations yield the system

$$\begin{array}{ccc} 2x + ay = dx + 2y & & (2-d)x + (2-a)y = 0 \\ ax + 2by = -2cx - dy & \text{or} & (a+2c)x + (2b+d)y = 0. \end{array}$$

The system holds for $z = x + iy$ whenever $2-d=0$, $2-a=0$, $a+2c=0$, and $2b+d=0$. That is, f is analytic for all z when $a=2$, $b=-1$, $c=-1$, and $d=2$.

17. $u = x^2 + y^2$, $v = 2xy$; $\frac{\partial u}{\partial x} = 2x$, $\frac{\partial v}{\partial y} = 2x$; $\frac{\partial u}{\partial y} = 2y$, $-\frac{\partial v}{\partial x} = -2y$

u and v are continuous and have continuous first partial derivatives. The Cauchy-Riemann equations are satisfied for any x and for $y=0$, that is, for points on the real axis. The function f is differentiable but not analytic along this axis; there is no neighborhood about any point $z=x$ within which f is differentiable.

18. $u = 3x^2y^2$, $v = -6x^2y^2$; $\frac{\partial u}{\partial x} = 6xy^2$, $\frac{\partial v}{\partial y} = -12x^2y$; $\frac{\partial u}{\partial y} = 6x^2y$, $-\frac{\partial v}{\partial x} = 12xy^2$

u and v are continuous and have continuous first partial derivatives. The Cauchy-Riemann equations are satisfied whenever $6xy(y+2x)=0$ and $6xy(x-2y)=0$. The point satisfying $y+2x=0$ and $x-2y=0$ is $z=0$. The points that satisfy $6xy=0$ are the points along the y -axis ($x=0$) or along the x -axis ($y=0$). The function f is differentiable but not analytic on either axis; there is no neighborhood about any point $z=x$ or $z=iy$ within which f is differentiable.

19. $u = x^3 + 3xy^2 - x$, $v = y^3 + 3x^2y - y$; $\frac{\partial u}{\partial x} = 3x^2 + 3y^2 - 1$, $\frac{\partial v}{\partial y} = 3y^2 + 3x^2 - 1$; $\frac{\partial u}{\partial y} = 6xy$, $-\frac{\partial v}{\partial x} = -6xy$.

u and v are continuous and have continuous first partial derivatives. The Cauchy-Riemann equations are satisfied whenever $6xy = -6xy$ or $12xy = 0$. The points satisfying $12xy = 0$ are the points along the y -axis ($x=0$) or along the x -axis ($y=0$). The function f is differentiable but not analytic on either axis; there is no neighborhood about any point $z=x$ or $z=iy$ within which f is differentiable.

20. $u = x^2 - x + y$, $v = y^2 - 5y - x$; $\frac{\partial u}{\partial x} = 2x - 1$, $\frac{\partial v}{\partial y} = 2y - 5$; $\frac{\partial u}{\partial y} = 1$, $-\frac{\partial v}{\partial x} = 1$

u and v are continuous and have continuous first partial derivatives. The Cauchy-Riemann equations are satisfied whenever $2x-1=2y-5$ or for points on the line $y=x+2$. The function f is differentiable but not analytic on this line; there is no neighborhood about any point $z=x+(x+2)i$ within which f is differentiable.

21. Since f is entire,

$$f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = e^x \cos y + ie^x \sin y = f(z).$$

22. Since f is entire,

$$f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = -2ye^{x^2-y^2} \sin 2xy + 2xe^{x^2-y^2} \cos 2xy + i(2ye^{x^2-y^2} \cos 2xy + 2xe^{x^2-y^2} \sin 2xy).$$

23. $\frac{\partial^2 u}{\partial x^2} = 0$, $\frac{\partial^2 u}{\partial y^2} = 0$ gives $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$. Thus u is harmonic. Now $\frac{\partial u}{\partial x} = 1 = \frac{\partial v}{\partial y}$ implies $v = y + h(x)$, $\frac{\partial u}{\partial y} = 0 = -\frac{\partial v}{\partial x}$ implies $0 = -h'(x)$, and so $h(x) = C$ (a constant.) Therefore $f(z) = x + i(y + C)$.

24. $\frac{\partial^2 u}{\partial x^2} = 0$, $\frac{\partial^2 u}{\partial y^2} = 0$ gives $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$. Thus u is harmonic. Now $\frac{\partial u}{\partial x} = 2-2y = \frac{\partial v}{\partial y}$ implies $v = 2y - y^2 + h(x)$, $\frac{\partial u}{\partial y} = -2x = -\frac{\partial v}{\partial x} = -h'(x)$ implies $h'(x) = 2x$ or $h(x) = x^2 + C$. Therefore $f(z) = 2x - 2xy + i(2y - y^2 + x^2 + C)$.

17.5 Cauchy-Riemann Equations

25. $\frac{\partial^2 u}{\partial x^2} = 2$, $\frac{\partial^2 u}{\partial y^2} = -2$ gives $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$. Thus u is harmonic. Now $\frac{\partial u}{\partial x} = 2x = \frac{\partial v}{\partial y}$ implies $v = 2xy + h(x)$,

$$\frac{\partial u}{\partial y} = -2y = -\frac{\partial v}{\partial x} = -2y - h'(x) \text{ implies } h'(x) = 0 \text{ or } h(x) = C. \text{ Therefore } f(z) = x^2 - y^2 + i(2xy + C).$$

26. $\frac{\partial^2 u}{\partial x^2} = -24xy$, $\frac{\partial^2 u}{\partial y^2} = 24xy$ gives $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$. Thus u is harmonic. Now $\frac{\partial u}{\partial x} = 4y^3 - 12x^2y + 1 = \frac{\partial v}{\partial y}$ implies $v = y^4 - 6x^2y^2 + y + h(x)$, $\frac{\partial u}{\partial y} = 12xy^2 - 4x^3 = -\frac{\partial v}{\partial x} = 12xy^2 - h'(x)$ implies $h'(x) = 4x^3$ or $h(x) = x^4 + C$.

$$\text{Therefore } f(z) = 4xy^3 - 4x^3y + x + i(y^4 - 6x^2y^2 + y + x^4 + C).$$

27. $\frac{\partial^2 u}{\partial x^2} = \frac{2y^2 - 2x^2}{(x^2 + y^2)^2}$, $\frac{\partial^2 u}{\partial y^2} = \frac{2x^2 - 2y^2}{(x^2 + y^2)^2}$ gives $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$. Thus u is harmonic. Now $\frac{\partial u}{\partial x} = \frac{2x}{x^2 + y^2} = \frac{\partial v}{\partial y}$

$$\text{implies } v = 2 \tan^{-1} \frac{y}{x} + h(x), \quad \frac{\partial u}{\partial y} = \frac{2y}{x^2 + y^2} = -\frac{\partial v}{\partial x} = \frac{2y}{x^2 + y^2} - h'(x) \text{ implies } h'(x) = 0 \text{ or } h(x) = C.$$

$$\text{Therefore } f(z) = \log_e(x^2 + y^2) + i \left(\tan^{-1} \frac{y}{x} + C \right), \quad z \neq 0.$$

28. $\frac{\partial^2 u}{\partial x^2} = 2e^x \cos y + e^x(x \cos y - y \sin y)$, $\frac{\partial^2 u}{\partial y^2} = e^x(-x \cos y + y \sin y - 2 \cos y)$ gives $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$. Thus u is

harmonic. Now $\frac{\partial u}{\partial x} = e^x \cos y + e^x(x \cos y - y \sin y) = \frac{\partial v}{\partial y}$. Integrating by parts with respect to y implies

$$v = e^x \sin y + e^x(x \sin y + y \cos y - \sin y) + h(x) = xe^x \sin y + ye^x \cos y + h(x),$$

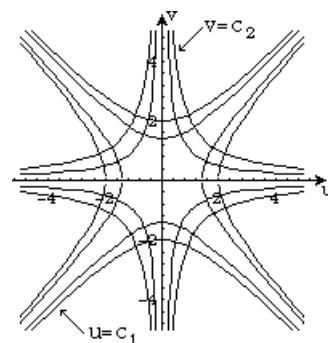
and

$$\frac{\partial u}{\partial y} = -xe^x \sin y - ye^x \cos y - e^x \sin y = -\frac{\partial v}{\partial x} = -xe^x \sin y - e^x \sin y - ye^x \cos y + h'(x)$$

implies $h'(x) = 0$ or $h(x) = C$. Therefore

$$f(z) = e^x(x \cos y - y \sin y) + ie^x(x \sin y + y \cos y + C).$$

29. The level curves $u(x, y) = c_1$ and $v(x, y) = c_2$ are the families of hyperbolas $x^2 - y^2 = c_1$ and $2xy = c_2$, respectively. The graphs of these families are displayed on the same axes in the figure.



30. $f(z) = \frac{x}{x^2 + y^2} - i \frac{y}{x^2 + y^2}$. The level curves $u(x, y) = c_1$ and $v(x, y) = c_2$ are the family of circles $x = c_1(x^2 + y^2)$ and $-y = c_2(x^2 + y^2)$, with the exception that $(0, 0)$ is not on the circumference of any circle.

31. $f(z) = x + \frac{x}{x^2 + y^2} + i \left(y - \frac{y}{x^2 + y^2} \right)$. The level curve $v(x, y) = 0$ is described by $y - \frac{y}{x^2 + y^2} = 0$ or $y(x^2 + y^2 - 1) = 0$. We see that either $y = 0$ or $x^2 + y^2 = 1$. Thus $v(x, y) = 0$ gives either the x -axis (without the origin $(0, 0)$) or the unit circle $x^2 + y^2 = 1$.

32. If $\nabla u = \frac{\partial u}{\partial x} \mathbf{i} + \frac{\partial u}{\partial y} \mathbf{j}$ and $\nabla v = \frac{\partial v}{\partial x} \mathbf{i} + \frac{\partial v}{\partial y} \mathbf{j}$, then $\nabla u \cdot \nabla v = \frac{\partial u}{\partial x} \frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \frac{\partial v}{\partial y}$. By the Cauchy-Riemann equations this becomes

$$\nabla u \cdot \nabla v = \frac{\partial v}{\partial y} \frac{\partial v}{\partial x} + \left(-\frac{\partial v}{\partial x}\right) \frac{\partial v}{\partial y} = 0.$$

Since the gradients of u and v are orthogonal vectors, the level curves $u(x, y) = c_1$ and $u(x, y) = c_2$ are orthogonal families.

EXERCISES 17.6

Exponential and Logarithmic Functions

1. $e^{\frac{\pi}{6}i} = \cos \frac{\pi}{6} + i \sin \frac{\pi}{6} = \frac{\sqrt{3}}{2} + \frac{1}{2}i$
2. $e^{-\frac{\pi}{3}i} = \cos \frac{\pi}{3} - i \sin \frac{\pi}{3} = \frac{1}{2} - \frac{\sqrt{3}}{2}i$
3. $e^{-1+\frac{\pi}{4}i} = e^{-1} \cos \frac{\pi}{4} + ie^{-1} \sin \frac{\pi}{4} = e^{-1} \left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i \right)$
4. $e^{2-\frac{\pi}{2}i} = e^2 \cos \left(-\frac{\pi}{2}\right) + ie^2 \sin \left(-\frac{\pi}{2}\right) = -e^2i$
5. $e^{\pi+\pi i} = e^{\pi} \cos \pi + ie^{\pi} \sin \pi = -e^{\pi}$
6. $e^{-\pi+\frac{3\pi}{2}i} = e^{-\pi} \cos \frac{3\pi}{2} + ie^{-\pi} \sin \frac{3\pi}{2} = -e^{-\pi}i$
7. $e^{1.5+2i} = e^{1.5} \cos 2 + ie^{1.5} \sin 2 = -1.8650 + 4.0752i$
8. $e^{-0.3+0.5i} = e^{-0.3} \cos 0.5 + ie^{-0.3} \sin 0.5 = 0.6501 + 0.3552i$
9. $e^{5i} = \cos 5 + i \sin 5 = 0.2837 - 0.9589i$
10. $e^{-0.23-i} = e^{-0.23} \cos(-1) + ie^{-0.23} \sin(-1) = 0.4293 - 0.6686i$
11. $e^{\frac{11\pi}{12}i} = \cos \frac{11\pi}{12} + i \sin \frac{11\pi}{12} = -0.9659 + 0.2588i$
12. $e^{5+\frac{5\pi}{2}i} = e^5 \cos \frac{5\pi}{2} + ie^5 \sin \frac{5\pi}{2} = e^5i$
13. $e^{-iz} = e^{y-xi} = e^y \cos x - ie^y \sin x$
14. $e^{2\bar{z}} = e^{2x-2yi} = e^{2x} \cos 2y - ie^{2x} \sin 2y$
15. $e^{z^2} = e^{x^2-y^2+2xyi} = e^{x^2-y^2} \cos 2xy + ie^{x^2-y^2} \sin 2xy$
16. $e^{1/z} = e^{x/(x^2+y^2)-iy/(x^2+y^2)} = e^{x/(x^2+y^2)} \cos \frac{y}{x^2+y^2} - ie^{x/(x^2+y^2)} \sin \frac{y}{x^2+y^2}$
17. $|e^z|^2 = e^{2x} \cos^2 y + e^{2x} \sin^2 y = e^{2x}(\cos^2 y + \sin^2 y) = e^{2x}$ implies $|e^z| = e^x$.
18. $\frac{e^{z_1}}{e^{z_2}} = \frac{e^{x_1} \cos y_1 + ie^{x_1} \sin y_1}{e^{x_2} \cos y_2 + ie^{x_2} \sin y_2} = \frac{(e^{x_1} \cos y_1 + ie^{x_1} \sin y_1)(e^{x_2} \cos y_2 - ie^{x_2} \sin y_2)}{e^{2x_2}}$
 $= e^{x_1-x_2}[(\cos y_1 \cos y_2 + \sin y_1 \sin y_2) + i(\sin y_1 \cos y_2 - \cos y_1 \sin y_2)]$
 $= e^{x_1-x_2}[\cos(y_1 - y_2) + i \sin(y_1 - y_2)] = e^{x_1-x_2+i(y_1-y_2)} = e^{(x_1+iy_1)-(x_2+iy_2)} = e^{z_1-z_2}$

17.6 Exponential and Logarithmic Functions

$$19. e^{z+\pi i} = e^{x+(y+\pi)i} = e^x[\cos(y+\pi) + i\sin(y+\pi)] = e^x[\cos(y-\pi) + i\sin(y-\pi)] = e^{x+(y-\pi)i} = e^{z-\pi i}$$

$$20. (e^z)^n = (e^x[\cos y + i\sin y])^n = e^{nx}[\cos y + i\sin y]^n = e^{nx}[\cos ny + i\sin ny] = e^{nz}, \quad n \text{ an integer}$$

$$21. u = e^x \cos y, \quad v = -e^x \sin y; \quad \frac{\partial u}{\partial x} = e^x \cos y, \quad \frac{\partial v}{\partial y} = -e^x \cos y; \quad \frac{\partial u}{\partial y} = -e^x \sin y, \quad -\frac{\partial v}{\partial x} = e^x \sin y$$

Since the Cauchy-Riemann equations are not satisfied at any point, f is nowhere analytic.

$$22. (a) \quad u = e^{x^2-y^2} \cos 2xy, \quad v = e^{x^2-y^2} \sin 2xy; \quad \frac{\partial u}{\partial x} = -2ye^{x^2-y^2} \sin 2xy + 2xe^{x^2-y^2} \cos 2xy = \frac{\partial v}{\partial y};$$

$$\frac{\partial u}{\partial y} = -2xe^{x^2-y^2} \sin 2xy - 2ye^{x^2-y^2} \cos 2xy = -\frac{\partial v}{\partial x}$$

Since u , v , and their first partial derivatives are continuous, and u and v satisfy the Cauchy-Riemann equations everywhere, the function f is differentiable everywhere. Hence f is entire.

$$(b) \quad \frac{\partial^2 u}{\partial x^2} = -4y^2 e^{x^2-y^2} \cos 2xy - 4xy e^{x^2-y^2} \sin 2xy - 4xy e^{x^2-y^2} \sin 2xy + \cos 2xy [4x^2 e^{x^2-y^2} + 2e^{x^2-y^2}];$$

$$\frac{\partial^2 u}{\partial y^2} = -4x^2 e^{x^2-y^2} \cos 2xy + 4xy e^{x^2-y^2} \sin 2xy + 4xy e^{x^2-y^2} \sin 2xy + \cos 2xy [4y^2 e^{x^2-y^2} - 2e^{x^2-y^2}]$$

Since $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ the function u is harmonic throughout the z -plane.

$$23. \ln(-5) = \log_e 5 + i(\pi + 2n\pi) = 1.16094 + (\pi + 2n\pi)i$$

$$24. \ln(-ei) = \log_e e + i\left(-\frac{\pi}{2} + 2n\pi\right) = 1 + \left(-\frac{\pi}{2} + 2n\pi\right)i$$

$$25. \ln(-2 + 2i) = \log_e 2\sqrt{2} + i\left(\frac{3\pi}{4} + 2n\pi\right) = 1.0397 + \left(\frac{3\pi}{4} + 2n\pi\right)i$$

$$26. \ln(1 + i) = \log_e \sqrt{2} + i\left(\frac{\pi}{4} + 2n\pi\right) = 0.3466 + \left(\frac{\pi}{4} + 2n\pi\right)i$$

$$27. \ln(\sqrt{2} + \sqrt{6}i) = \log_e 2\sqrt{2} + i\left(\frac{\pi}{3} + 2n\pi\right) = 1.0397 + \left(\frac{\pi}{3} + 2n\pi\right)i$$

$$28. \ln(-\sqrt{3} + i) = \log_e 2 + i\left(\frac{5\pi}{6} + 2n\pi\right) = 0.6932 + \left(\frac{5\pi}{6} + 2n\pi\right)i$$

$$29. \operatorname{Ln}(6 - 6i) = \log_e 6\sqrt{2} + i\left(-\frac{\pi}{4}\right) = 2.1383 - \frac{\pi}{4}i$$

$$30. \operatorname{Ln}(-e^3) = \log_e e^3 + \pi i = 3 + \pi i$$

$$31. \operatorname{Ln}(-12 + 5i) = \log_e 13 + i\left(\tan^{-1}\left(-\frac{5}{12}\right) + \pi\right) = 2.5649 + 2.7468i$$

$$32. \operatorname{Ln}(3 - 4i) = \log_e 5 + i\tan^{-1}\left(-\frac{4}{3}\right) = 1.6094 - 0.9273i$$

$$33. \operatorname{Ln}(1 + \sqrt{3}i)^5 = \operatorname{Ln}(16 - 16\sqrt{3}i) = \log_e 32 - \frac{\pi}{3}i = 3.4657 - \frac{\pi}{3}i$$

$$34. \operatorname{Ln}(1 + i)^4 = \operatorname{Ln}(-4) = \log_e 4 + \pi i = 1.3863 + \pi i$$

$$35. z = \ln(4i) = \log_e 4 + i\left(\frac{\pi}{2} + 2n\pi\right) = 1.3863 + \left(\frac{\pi}{2} + 2n\pi\right)i$$

$$36. \frac{1}{z} = \ln(-1) = \log_e 1 + i(\pi + 2n\pi) = (2n+1)\pi i \quad \text{and so} \quad z = -\frac{i}{(2n+1)\pi}.$$

$$37. z - 1 = \ln(-ie^2) = \log_e e^2 + i\left(\frac{3\pi}{2} + 2n\pi\right) = 2 + \left(\frac{3\pi}{2} + 2n\pi\right)i \quad \text{and so} \quad z = 3 + \left(\frac{3\pi}{2} + 2n\pi\right)i.$$

38. By the quadratic formula, $e^z = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$ or $e^z = -\frac{1}{2} - \frac{\sqrt{3}}{2}i$. Hence

$$z = \ln \left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i \right) = \left(\frac{2\pi}{3} + 2n\pi \right) i \quad \text{or} \quad z = \ln \left(-\frac{1}{2} - \frac{\sqrt{3}}{2}i \right) = \left(\frac{4\pi}{3} + 2n\pi \right) i.$$

39. $(-i)^{4i} = e^{4i \ln(-i)} = e^{4i[\log_e 1 + i(-\frac{\pi}{2} + 2n\pi)]} = e^{(2-8n)\pi}$

40. $3^{1/\pi} = e^{\frac{1}{\pi} \ln 3} = e^{\frac{1}{\pi}[\log_e 3 + 2n\pi i]} = e^{-2n} \left[\cos \left(\frac{1}{\pi} \log_e 3 \right) + i \sin \left(\frac{1}{\pi} \log_e 3 \right) \right] = e^{-2n}[0.9395 + 0.3426i]$

41. $(1+i)^{(1+i)} = e^{(1+i) \ln(1+i)} = e^{(1+i)[\log_e \sqrt{2} + i(\frac{\pi}{4} + 2n\pi)]}$
 $= e^{\log_e \sqrt{2} - (\frac{\pi}{4} + 2n\pi)} \left[\cos \left(\frac{\pi}{4} + \log_e \sqrt{2} \right) + i \sin \left(\frac{\pi}{4} + \log_e \sqrt{2} \right) \right] = e^{-2n\pi}[0.2740 + 0.5837i]$

42. $(1-i)^{2i} = e^{2i \ln(1-i)} = e^{2i[\log_e \sqrt{2} + i(-\frac{\pi}{4} + 2n\pi)]} = e^{\frac{\pi}{2} - 4n\pi} [\cos(\log_e 2) + i \sin(\log_e 2)] = e^{-4n\pi}[3.7004 + 3.0737i]$

43. $(-1)^{-\frac{2i}{\pi}} = e^{-\frac{2i}{\pi} \ln(-1)} = e^{-\frac{2i}{\pi}(\pi i)} = e^2 = 7.3891$

44. $(1-i)^{2i} = e^{2i \ln(1-i)} = e^{2i[\log_e \sqrt{2} - \frac{\pi}{4}i]} = e^{\frac{\pi}{2}} [\cos(\log_e 2) + i \sin(\log_e 2)] = 3.7004 + 3.0737i$

45. If $z_1 = i$ and $z_2 = -1 + i$ then

$$\text{Ln}(z_1 z_2) = \text{Ln}(-1 - i) = \log_e \sqrt{2} - \frac{3\pi}{4}i,$$

whereas

$$\text{Ln} z_1 + \text{Ln} z_2 = \frac{\pi}{2}i + \left(\log_e \sqrt{2} + \frac{3\pi}{4}i \right) = \log_e \sqrt{2} + \frac{5\pi}{4}i.$$

46. If $z_1 = -i$ and $z_2 = i$ then

$$\text{Ln}(z_1/z_2) = \text{Ln}(-1) = \pi i, \quad \text{whereas} \quad \text{Ln} z_1 - \text{Ln} z_2 = -\frac{\pi}{2}i - \frac{\pi}{2}i = -\pi i.$$

47. (a) The statement is false.

$$\text{Ln}(-1+i)^2 = \text{Ln}(-2i) = \log_e 2 - \frac{\pi}{2}i, \quad \text{whereas} \quad 2\text{Ln}(-1+i) = 2 \left(\log_e \sqrt{2} + \frac{3\pi}{4}i \right) = \log_e 2 + \frac{3\pi}{2}i.$$

(b) The statement is false.

$$\text{Ln} i^3 = \text{Ln}(-i) = -\frac{\pi}{2}i, \quad \text{whereas} \quad 3\text{Ln} i = \frac{3\pi}{2}i.$$

(c) The statement is true. If we take $\arg(-i) = \frac{3\pi}{2}$ then $\ln i^3 = \ln(-i) = \frac{3\pi}{2}i$ for $n = 0$. Also, $3 \ln i = 3 \left(\frac{\pi}{2}i \right)$.

48. (a) $(i^i)^2 = (e^{i \ln i})^2 = [e^{-(\frac{\pi}{2} + 2n\pi)}]^2 = e^{-(\pi + 4n\pi)}$ and $i^{2i} = e^{2i \ln i} = e^{-(\pi + 4n\pi)}$

(b) $(i^2)^i = (-1)^i = e^{i \ln(-1)} = e^{-(\pi + 2n\pi)}$, whereas $i^{2i} = e^{-(\pi + 4n\pi)}$

49. Since $|z| = \sqrt{x^2 + y^2}$ and $\text{Arg} z = \tan^{-1} \frac{y}{x}$ for $x > 0$ we have

$$\text{Ln} z = \log_e |z| + i \text{Arg} z = \log_e (x^2 + y^2)^{1/2} + i \tan^{-1} \frac{y}{x} = \frac{1}{2} \log_e (x^2 + y^2) + i \tan^{-1} \frac{y}{x}.$$

50. (a) $u = \log_e (x^2 + y^2)$; $\frac{\partial^2 u}{\partial x^2} = \frac{2(y^2 - x^2)}{(x^2 + y^2)^2}$, $\frac{\partial^2 u}{\partial y^2} = \frac{2(x^2 - y^2)}{(x^2 + y^2)^2}$

Since $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ the function u is harmonic in any domain not containing the point $(0, 0)$.

(b) $v = \tan^{-1} \frac{y}{x}$; $\frac{\partial^2 v}{\partial x^2} = \frac{2xy}{(x^2 + y^2)^2}$, $\frac{\partial^2 v}{\partial y^2} = \frac{-2xy}{(x^2 + y^2)^2}$

Since $\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0$ the function v is harmonic in any domain not containing the point $(0, 0)$.

EXERCISES 17.7

Trigonometric and Hyperbolic Functions

1. $\cos(3i) = \cosh 3 = 10.0677$
2. $\sin(-2i) = i \sinh(-2) = -3.6269i$
3. $\sin\left(\frac{\pi}{4} + i\right) = \sin \frac{\pi}{4} \cosh(1) + i \cos \frac{\pi}{4} \sinh(1) = 1.0911 + 0.8310i$
4. $\cos(2 - 4i) = \cos(2) \cosh(-4) - \sin(2) \sinh(-4) = -11.3642 - 24.8147i$
5. $\tan(i) = \frac{\sin(i)}{\cos(i)} = \frac{i \sinh(1)}{\cosh(1)} = 0.7616i$
6. $\cot\left(\frac{\pi}{2} + 3i\right) = \frac{\cos(\frac{\pi}{2} + 3i)}{\sin(\frac{\pi}{2} + 3i)} = \frac{-i \sinh(3)}{\cosh(3)} = -0.9951i$
7. $\sec(\pi + i) = \frac{1}{\cos(\pi + i)} = \frac{1}{-\cosh(1)} = -0.6481$
8. $\csc(1 + i) = \frac{1}{\sin(1 + i)} = \frac{1}{\sin(1) \cosh(1) + i \cos(1) \sinh(1)} = 0.6215 - 0.3039i$
9. $\cosh(\pi i) = \cos(i(\pi i)) = \cos(-\pi) = \cos \pi = -1$
10. $\sinh\left(\frac{3\pi}{2} i\right) = -i \sin\left[i\left(\frac{3\pi}{2} i\right)\right] = -i \sin\left(-\frac{3\pi}{2}\right) = i \sin \frac{3\pi}{2} = -i$
11. $\sinh\left(1 + \frac{\pi}{3} i\right) = \sinh(1) \cos \frac{\pi}{3} + i \cosh(1) \sin \frac{\pi}{3} = 0.5876 + 1.3363i$
12. $\cosh(2 + 3i) = \cosh(2) \cos(3) + i \sinh(2) \sin(3) = -3.7245 + 0.5118i$
13. $\sin\left(\frac{\pi}{2} + i \ln 2\right) = \sin \frac{\pi}{2} \cosh(\ln 2) + i \cos \frac{\pi}{2} \sinh(\ln 2) = \frac{e^{\ln 2} + e^{\ln 2^{-1}}}{2} = \frac{2 + \frac{1}{2}}{2} = \frac{5}{4}$
14. $\cos\left(\frac{\pi}{2} + i \ln 2\right) = \cos \frac{\pi}{2} \cosh(\ln 2) - i \sin \frac{\pi}{2} \sinh(\ln 2) = -i \cdot \frac{e^{\ln 2} - e^{\ln 2^{-1}}}{2} = -i \cdot \frac{2 - \frac{1}{2}}{2} = -\frac{3}{4}i$
15. $\frac{e^{iz} - e^{-iz}}{2i} = 2$ gives $e^{2(iz)} - 4ie^{iz} - 1 = 0$. By the quadratic formula, $e^{iz} = 2i \pm \sqrt{3}i$ and so

$$iz = \ln[(2 \pm \sqrt{3})i]$$

$$z = -i \left[\log_e(2 \pm \sqrt{3}) + \left(\frac{\pi}{2} + 2n\pi\right)i \right] = \frac{\pi}{2} + 2n\pi - i \log_e(2 \pm \sqrt{3}), \quad n = 0, \pm 1, \pm 2, \dots$$

16. $\frac{e^{iz} + e^{-iz}}{2} = -3i$ gives $e^{2(iz)} + 6ie^{iz} + 1 = 0$. By the quadratic formula, $e^{iz} = -3i \pm \sqrt{10}i$ and so
 $iz = \ln[-3 \pm \sqrt{10}]i$. Hence

$$z = -i \left[\log_e(\sqrt{10} - 3) + \left(\frac{\pi}{2} + 2n\pi\right)i \right] \quad \text{or} \quad z = -i \left[\log_e(\sqrt{10} + 3) + \left(\frac{3\pi}{2} + 2n\pi\right)i \right]$$

$$z = \frac{\pi}{2} + 2n\pi - i \log_e(\sqrt{10} - 3) \quad \text{or} \quad z = \frac{3\pi}{2} + 2n\pi - i \log_e(\sqrt{10} + 3)$$

$$n = 0, \pm 1, \pm 2, \dots$$

17. $\frac{e^z - e^{-z}}{2} = i$ gives $e^{2z} - 2ie^z - 1 = 0$. By the quadratic formula, $e^z = -i$ and so

$$z = \ln(-i) = \log_e 1 + \left(-\frac{\pi}{2} + 2n\pi\right)i = \left(-\frac{\pi}{2} + 2n\pi\right)i, \quad n = 0, \pm 1, \pm 2, \dots$$

18. $\frac{e^z - e^{-z}}{2} = -1$ gives $e^{2z} + 2e^z - 1 = 0$. By the quadratic formula, $e^z = -1 \pm \sqrt{2}$, and so

$$z = \ln(-1 \pm \sqrt{2})$$

$$z = \log_e(\sqrt{2} - 1) + 2n\pi i \quad \text{or} \quad z = \log_e(\sqrt{2} + 1) + (\pi + 2n\pi)i,$$

$$n = 0, \pm 1, \pm 2, \dots$$

19. $\cos z = \sin z$ gives $\tan z = 1$. One solution is $z = \frac{\pi}{4}$. Since $\tan z$ is π -periodic, $z = \frac{\pi}{4} + n\pi$, $n = 0, \pm 1, \pm 2, \dots$ are also solutions. That these are the only solutions can be proved by solving

$$\frac{e^{iz} + e^{-iz}}{2} = \frac{e^{iz} - e^{-iz}}{2i}$$

by the method illustrated in Problems 15-18.

20. $\cos z = i \sin z$ gives $e^{iz} + e^{-iz} = e^{iz} - e^{-iz}$ or $e^{-iz} = 0$. Since this last equation has no solutions, the original equation has no solutions.
21. $\cos z = \cosh 2$ implies $\cos x \cosh y - i \sin x \sinh y = \cosh 2 + 0i$ and so we must have $\cos x \cosh y = \cosh 2$ and $\sin x \sinh y = 0$. The last equation has solutions $x = n\pi$, $n = 0, \pm 1, \pm 2, \dots$, or $y = 0$. For $y = 0$ the first equation becomes $\cos x = \cosh 2$. Since $\cosh 2 > 1$ this equation has no solutions. For $x = n\pi$ the first equation becomes $(-1)^n \cosh y = \cosh 2$. Since $\cosh y > 0$ we see n must be even, say, $n = 2k$, $k = 0, \pm 1, \pm 2, \dots$. Now $\cosh y = \cosh 2$ implies $y = \pm 2$. Solutions of the original equation are then

$$z = 2k\pi \pm 2i, \quad k = 0, \pm 1, \pm 2, \dots$$

22. $\sin z = i \sinh 2$ implies $\sin x \cosh y + i \cos x \sinh y = 0 + i \sinh 2$ and so we must have $\sin x \cosh y = 0$ and $\cos x \sinh y = \sinh 2$. Since $\cosh y > 0$ for all real numbers, the first equation has only the solutions $x = n\pi$, $n = 0, \pm 1, \pm 2, \dots$. For $x = n\pi$ the second equation becomes $(-1)^n \sinh y = \sinh 2$. If n is even, $\sinh y = \sinh 2$ implies $y = 2$ ($\sinh y$ is one-to-one.) If n is odd, $-\sinh y = \sinh 2$ implies $\sinh y = -\sinh(-2)$ and so $y = -2$. Solutions of the original equation are then

$$z = 2k\pi + 2i, \quad z = (2k + 1)\pi - 2i, \quad k = 0, \pm 1, \pm 2, \dots$$

$$\begin{aligned} 23. \quad \cos z &= \frac{e^{i(x+iy)} + e^{-i(x+iy)}}{2} = \frac{1}{2}(e^{-y}e^{ix} + e^ye^{-ix}) = \frac{1}{2}[e^{-y}(\cos x + i \sin x) + e^y(\cos x - i \sin x)] \\ &= \cos x \left(\frac{e^y + e^{-y}}{2}\right) - i \sin x \left(\frac{e^y - e^{-y}}{2}\right) = \cos x \cosh y - i \sin x \sinh y \end{aligned}$$

$$\begin{aligned} 24. \quad \sinh z &= \frac{e^{x+iy} - e^{-x-iy}}{2} = \frac{1}{2}(e^xe^{iy} - e^{-x}e^{-iy}) = \frac{1}{2}[e^x(\cos y + i \sin y) - e^{-x}(\cos y - i \sin y)] \\ &= \left(\frac{e^x - e^{-x}}{2}\right) \cos y + i \left(\frac{e^x + e^{-x}}{2}\right) \sin y = \sinh x \cos y + i \cosh x \sin y \end{aligned}$$

$$\begin{aligned} 25. \quad \cosh z &= \frac{e^{x+iy} + e^{-x-iy}}{2} = \frac{1}{2}(e^xe^{iy} + e^{-x}e^{-iy}) = \frac{1}{2}[e^x(\cos y + i \sin y) + e^{-x}(\cos y - i \sin y)] \\ &= \left(\frac{e^x + e^{-x}}{2}\right) \cos y + i \left(\frac{e^x - e^{-x}}{2}\right) \sin y = \cosh x \cos y + i \sinh x \sin y \end{aligned}$$

17.7 Trigonometric and Hyperbolic Functions

26. $|\sinh z|^2 = \sinh^2 x \cos^2 y + \cosh^2 x \sin^2 y = \sinh^2 x \cos^2 y + (1 + \sinh^2 x) \sin^2 y$
 $= \sinh^2 x (\cos^2 y + \sin^2 y) + \sin^2 y = \sinh^2 x + \sin^2 y$
27. $|\cosh z|^2 = \cosh^2 x \cos^2 y + \sinh^2 x \sin^2 y = (1 + \sinh^2 x) \cos^2 y + \sinh^2 x \sin^2 y$
 $= \cos^2 y + \sinh^2 x (\cos^2 y + \sin^2 y) = \cos^2 y + \sinh^2 x$
28. $\cos^2 z + \sin^2 z = \left(\frac{e^{1z} + e^{-iz}}{2} \right)^2 + \left(\frac{e^{iz} - e^{-iz}}{2i} \right)^2 = \frac{1}{4} [e^{2iz} + 2 + e^{-2iz} - (e^{2iz} - 2 + e^{-2iz})] = \frac{4}{4} = 1$
29. $\cosh^2 z - \sinh^2 z = \left(\frac{e^z + e^{-z}}{2} \right)^2 - \left(\frac{e^z - e^{-z}}{2} \right)^2 = \frac{1}{4} [e^{2z} + 2 + e^{-2z} - (e^{2z} - 2 + e^{-2z})] = \frac{4}{4} = 1$
30. $\tan z = \frac{\sin z}{\cos z} = \frac{\sin z \cos z}{|\cos z|^2} = \frac{[\sin x \cosh y + i \cos x \sinh y][\cos x \cosh y + i \sin x \sinh y]}{\cos^2 x + \sinh^2 y}$
 $= \frac{(\sin x \cos x \cosh^2 y - \sin x \cos x \sinh^2 y)}{\cos^2 x + \sinh^2 y} + i \frac{\cos^2 x \sinh y \cosh y + \sin^2 x \sinh y \cosh y}{\cos^2 x + \sinh^2 y}$
 $= \frac{\sin x \cos x (\cosh^2 y - \sinh^2 y)}{\cos^2 x + \sinh^2 y} + i \frac{\sin y \cosh y (\cos^2 x + \sin^2 x)}{\cos^2 x + \sinh^2 y}$
 $= \frac{\sin x \cos x}{\cos^2 x + \sinh^2 y} + i \frac{\sinh y \cosh y}{\cos^2 x + \sinh^2 y} = \frac{\sin 2x}{2(\cos^2 x + \sinh^2 y)} + i \frac{\sinh 2y}{2(\cos^2 x + \sinh^2 y)}$

But

$$2 \cos^2 x + 2 \sinh^2 y = (2 \cos^2 x - 1) + (2 \sinh^2 y + 1) = \cos 2x + \cosh 2y.$$

Therefore $\tan z = u + iz$ where

$$u = \frac{\sin 2x}{\cos 2x + \cosh 2y}, \quad v = \frac{\sinh 2y}{\cos 2x + \cosh 2y}.$$

31. $\tanh(z + \pi i) = \frac{\sinh(x + (y + \pi)i)}{\cosh(x + (y + \pi)i)} = \frac{\sinh x \cos(y + \pi) + i \cosh x \sin(y + \pi)}{\cosh x \cos(y + \pi) + i \sinh x \sin(y + \pi)}$
 $= \frac{-[\sinh x \cos y + i \cosh x \sin y]}{-[\cosh x \cos y + i \sinh x \sin y]} = \frac{-\sinh z}{-\cosh z} = \tanh z$
32. (a) $\overline{\sin z} = \sin x \cosh y - i \cos x \sinh y = \sin x \cosh(-y) + i \cos x \sinh(-y) = \sin(x - iy) = \sin \bar{z}$
(b) $\overline{\cos z} = \cos x \cosh y + i \sin x \sinh y = \cos x \cosh(-y) - i \sin x \sinh(-y) = \cos(x - iy) = \cos \bar{z}$

EXERCISES 17.8

Inverse Trigonometric and Hyperbolic Functions

1. $\sin^{-1}(-i) = -i \ln(1 \pm \sqrt{2}) = \begin{cases} 2n\pi - i \log_e(1 + \sqrt{2}) \\ (2n + 1)\pi - i \log_e(\sqrt{2} - 1) \end{cases}$

Since $\sqrt{2} - 1 = 1/(\sqrt{2} + 1)$ we can have

$$\sin^{-1}(-i) = \begin{cases} 2n\pi - i \log_e(1 + \sqrt{2}) \\ (2n + 1)\pi + i \log_e(1 + \sqrt{2}). \end{cases}$$

This can be written compactly as

$$\sin^{-1}(-i) = n\pi + (-1)^{n+1}i \log_e(1 + \sqrt{2}), \quad k = 0, \pm 1, \pm 2, \dots$$

$$2. \sin^{-1} \sqrt{2} = -i \ln[\sqrt{2} \pm 1]i = 2n\pi + \frac{\pi}{2} - i \log_e(\sqrt{2} \pm 1) = 2n\pi + \frac{\pi}{2} \pm i \log_e(1 + \sqrt{2}), \quad n = 0, \pm 1, \pm 2, \dots$$

$$3. \sin^{-1} 0 = -i \ln(\pm 1) = \begin{cases} 2n\pi + i \log_e 1 \\ (2n+1)\pi + i \log_e 1 \end{cases} = \begin{cases} 2n\pi \\ (2n+1)\pi \end{cases} = n\pi, \quad n = 0, \pm 1, \pm 2, \dots$$

$$4. \sin^{-1} \frac{13}{5} = -i \ln \left[\left(\frac{13}{5} \pm \frac{12}{5} \right) i \right] = \begin{cases} 2n\pi + \frac{\pi}{2} - i \log_e 5 \\ 2n\pi + \frac{\pi}{2} - i \log_e \frac{1}{5} \end{cases} = 2n\pi + \frac{\pi}{2} \pm i \log_e 5, \quad n = 0, \pm 1, \pm 2, \dots$$

$$5. \cos^{-1} 2 = -i \ln(2 \pm \sqrt{3}) = \begin{cases} 2n\pi - i \log_e(2 + \sqrt{3}) \\ 2n\pi - i \log_e(2 - \sqrt{3}) \end{cases}$$

Since $2 - \sqrt{3} = 1/(2 + \sqrt{3})$ this can be written compactly as

$$\cos^{-1} 2 = 2n\pi \pm i \log_e(2 + \sqrt{3}), \quad k = 0, \pm 1, \pm 2, \dots$$

$$6. \cos^{-1} 2i = -i \ln[(2 \pm \sqrt{5})i] = \begin{cases} 2n\pi - \frac{\pi}{2} + i \log_e(2 + \sqrt{5}) \\ 2n\pi + \frac{\pi}{2} - i \log_e(2 + \sqrt{5}) \end{cases}, \quad n = 0, \pm 1, \pm 2, \dots$$

$$7. \cos^{-1} \frac{1}{2} = -i \ln \left(\frac{1}{2} \pm \frac{\sqrt{3}}{2} i \right) = \begin{cases} 2n\pi + \frac{\pi}{3} - i \log_e 1 \\ 2n\pi + \frac{\pi}{3} - i \log_e 1 \end{cases} = 2n\pi \pm \frac{\pi}{2}, \quad n = 0, \pm 1, \pm 2, \dots$$

$$8. \cos^{-1} \frac{5}{3} = -i \ln \left(\frac{5}{3} \pm \frac{4}{3} \right) = \begin{cases} 2n\pi - i \log_e 3 \\ 2n\pi - i \log_e \frac{1}{3} \end{cases} = 2n\pi \pm i \log_e 3, \quad n = 0, \pm 1, \pm 2, \dots$$

$$9. \tan^{-1} 1 = \frac{i}{2} \ln \frac{i+1}{i-1} = \frac{i}{2} \ln(-i) = -n\pi + \frac{\pi}{4} + \frac{i}{2} \log_e 1 = \frac{\pi}{4} - n\pi, \quad n = 0, \pm 1, \pm 2, \dots$$

Note that this can also be written as $\tan^{-1} 1 = \frac{\pi}{4} + n\pi, \quad n = 0, \pm 1, \pm 2, \dots$

$$10. \tan^{-1} 3i = \frac{i}{2} \ln \left(\frac{4i}{-2i} \right) = \frac{i}{2} \ln(-2) = -\frac{\pi}{2} - n\pi + i \log_e \sqrt{2}, \quad n = 0, \pm 1, \pm 2, \dots$$

$$11. \sinh^{-1} \frac{4}{3} = \ln \left(\frac{4}{3} \pm \frac{5}{3} \right) = \begin{cases} \log_e 3 + 2n\pi i \\ \log_e \frac{1}{3} + (2n+1)\pi i \end{cases} = (-1)^n \log_e 3 + n\pi i, \quad n = 0, \pm 1, \pm 2, \dots$$

$$12. \cosh^{-1} i = \ln[(1 \pm \sqrt{2})i] = \begin{cases} \log_e(1 + \sqrt{2}) + (\frac{\pi}{2} + 2n\pi)i \\ \log_e(\sqrt{2} - 1) + (-\frac{\pi}{2} + 2n\pi)i \end{cases}, \quad n = 0, \pm 1, \pm 2, \dots$$

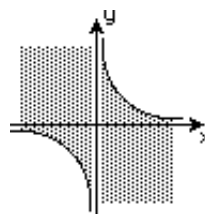
$$13. \tanh^{-1}(1 + 2i) = \frac{1}{2} \ln \frac{2+2i}{-2i} = \frac{1}{2} \ln(-1 + i) = \frac{1}{2} \left[\log_e \sqrt{2} + \left(\frac{3\pi}{4} + 2n\pi \right) i \right] = \frac{1}{4} \log_e 2 + \left(\frac{3\pi}{8} + n\pi \right) i$$

$$14. \tanh^{-1}(-\sqrt{3}i) = \frac{1}{2} \ln \frac{1-\sqrt{3}i}{1+\sqrt{3}i} = \frac{1}{2} \ln \left(-\frac{1}{2} - \frac{\sqrt{3}}{2} i \right) = \frac{1}{2} \left[\log_e 1 + \left(\frac{4\pi}{3} + 2n\pi \right) i \right] = \left(\frac{2\pi}{3} + n\pi \right) i,$$

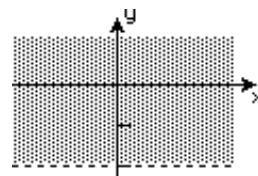
$n = 0, \pm 1, \pm 2, \dots$

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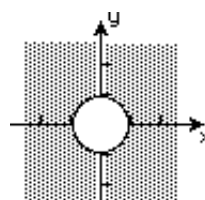
1. 0; 32
2. third
3. $-7/25$
4. $-8i$
5. $4/5$
6. The closed annular region between the circles $|z + 2| = 1$ and $|z + 2| = 3$. These circles have center at $z = -2$.
7. False. $\text{Arg}[(-1 + i) + (-1 - i)] = \text{Arg}(-2) = \pi$
8. $-5\pi/6$
9. $z = \ln(2i) = \log_e 2 + i \left(\frac{\pi}{2} + 2n\pi \right), \quad n = 0, \pm 1, \pm 2, \dots$
10. True
11. $(1 + i)^{2+i} = e^{(2+i)[\log_e 2 + \frac{\pi}{4}i]} = e^{(\log_e 2 - \frac{\pi}{4}) + i(\log_e \sqrt{2} + \frac{\pi}{2})} = e^{\log_e 2 - \frac{\pi}{4}} \left[\cos \left(\log_e \sqrt{2} + \frac{\pi}{2} \right) + i \sin \left(\log_e \sqrt{2} + \frac{\pi}{2} \right) \right]$
 $= -0.3097 + 0.8577i$
12. $f(-1 + i) = -33 + 26i$
13. False
14. $2\pi i$
15. $\text{Ln}(-ie^3) = \log_e e^3 + \left(-\frac{\pi}{2} \right) i = 3 - \frac{\pi}{2} i$
16. True
17. $58 - 4i$
18. $-\frac{1}{13} - \frac{17}{13}i$
19. $-8 + 8i$
20. $4e^{\pi i/12} = 4 \left(\cos \frac{\pi}{12} + i \sin \frac{\pi}{12} \right) = 3.8637 + 1.0353i$
21. The region satisfying $xy \leq 1$ is shown in the figure.



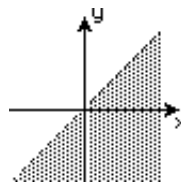
22. The region satisfying $y + 5 > 3$ or $y > -2$ is shown in the figure.



23. The region satisfying $|z| \geq 1$ is shown in the figure.



24. The region satisfying $y < x$ is shown in the figure.



25. Ellipse with foci $(0, -2)$ and $(0, 2)$

26. $\left| \frac{z-w}{1-z\bar{w}} \right|^2 = \frac{z-w}{1-z\bar{w}} \cdot \frac{\bar{z}-\bar{w}}{1-\bar{z}w} = \frac{z\bar{z} - z\bar{w} - w\bar{z} + w\bar{w}}{1 - \bar{z}w - z\bar{w} + z\bar{z}w\bar{w}} = \frac{1 - z\bar{w} - w\bar{z} + |w|^2}{1 - \bar{z}w - z\bar{w} + |w|^2} = 1,$
 since $|z|^2 = z\bar{z} = 1$ and $|w| \neq 1$.

27. The four fourth roots of $1 - i$ are given by

$$w_R = 2^{1/8} \left[\cos \left(-\frac{\pi}{16} + \frac{k\pi}{2} \right) + i \sin \left(-\frac{\pi}{16} + \frac{k\pi}{2} \right) \right], \quad n = 0, 1, 2, 3$$

$$w_0 = 2^{1/8} \left[\cos \left(-\frac{\pi}{16} \right) + i \sin \left(-\frac{\pi}{16} \right) \right] = 1.0696 - 0.2127i$$

$$w_1 = 2^{1/8} \left[\cos \frac{7\pi}{16} + i \sin \frac{7\pi}{16} \right] = 0.2127 + 1.0696i$$

$$w_2 = 2^{1/8} \left[\cos \frac{15\pi}{16} + i \sin \frac{15\pi}{16} \right] = -1.0696 + 0.2127i$$

$$w_3 = 2^{1/8} \left[\cos \frac{23\pi}{16} + i \sin \frac{23\pi}{16} \right] = -0.2127 - 1.0696i$$

28. $z^{3/2} = \frac{2}{5} + \frac{1}{5}i$ implies $z^3 = \frac{3}{25} + \frac{4}{25}i$. The three cube roots of $\frac{3}{25} + \frac{4}{25}i$ are

$$w_k = \left(\frac{1}{5} \right)^{1/3} \left[\cos \left(\frac{1}{3} \tan^{-1} \left(\frac{4}{3} \right) + \frac{2k\pi}{3} \right) + i \sin \left(\frac{1}{3} \tan^{-1} \left(\frac{4}{3} \right) + \frac{2k\pi}{3} \right) \right], \quad k = 0, 1, 2$$

$$w_0 = \left(\frac{1}{5} \right)^{1/3} [\cos(0.3091) + i \sin(0.3091)] = 0.5571 + 0.1779i$$

$$w_1 = \left(\frac{1}{5} \right)^{1/3} [\cos(2.4035) + i \sin(2.4035)] = -0.4326 + 0.3935i$$

$$w_2 = \left(\frac{1}{5} \right)^{1/3} [\cos(4.4979) + i \sin(4.4979)] = -0.1245 - 0.5714i.$$

29. Write $(1+i)/\sqrt{2} = e^{\pi i/4}$ so that

$$z^{24} = e^{6\pi i} = 1, \quad z^{20} = e^{5\pi i} = -1, \quad z^{12} = e^{3\pi i} = -1, \quad z^6 = e^{3\pi i/2} = -i.$$

Therefore

$$f \left(\frac{1+i}{\sqrt{2}} \right) = 1 - 3(-1) + 4(-1) - 5(-i) = 5i.$$

30. $\text{Im}(z - 3\bar{z}) = 4y$, $z\text{Re}(z^2) = (x^3 - xy^2) + i(x^2y - y^3)$. Thus,

$$f(z) = (4y + x^3 - xy^2 - 5x) + i(x^2y - y^3 - 5y).$$

31. $u = x^2 - y$, $v = y^2 - x$. When $x = 1$ we get the parametric equations $u = 1 - y$, $v = y^2 - 1$. Eliminating y then gives $v = (1 - u)^2 - 1 = u^2 - 2u$. This is an equation of a parabola.

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32. $u = x/(x^2 + y^2)$, $v = -y/(x^2 + y^2)$. When $x = 1$ we get the parametric equations $u = 1/(1 + y^2)$, $v = -y/(1 + y^2)$. From this we find $u^2 + v^2 - u = 0$. This describes a circle with the exception that $(0, 0)$ is not on its circumference.

33. $z = z^{-1}$ gives $z^2 = 1$ or $(z - 1)(z + 1) = 0$. Thus $z = \pm 1$.

34. $\bar{z} = 1/z$ gives $z\bar{z} = 1$ or $|z|^2 = 1$. All points on the circle $|z| = 1$ satisfy the equation.

35. $\bar{z} = -z$ gives $x = -x$ or $x = 0$. All complex numbers of the form $z = 0 + iy$ (pure imaginaries) satisfy the equation.

36. $z^2 = \bar{z}^2$ gives $xy = -xy$ or $xy = 0$. This implies $x = 0$ or $y = 0$. All real numbers ($y = 0$) and all pure imaginary numbers ($x = 0$) satisfy the equation.

37. $u = -2xy - 5x$, $v = x^2 - 5y - y^2$; $\frac{\partial u}{\partial x} = -2y - 5 = \frac{\partial v}{\partial y}$, $\frac{\partial u}{\partial y} = -2x = -\frac{\partial v}{\partial x}$; $f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = -2y - 5 + 2xi$

38. $u = x^3 + xy^2 - 4x$, $v = 4y - y^3 - x^2y$; $\frac{\partial u}{\partial x} = 3x^2 + y^2 - 4$, $\frac{\partial v}{\partial y} = 4 - 3y^2 - x^2$, $\frac{\partial u}{\partial y} = 2xy = -\frac{\partial v}{\partial x}$

The Cauchy-Riemann equations are satisfied at all points on the circle $x^2 + y^2 = 2$. Continuity of u , v , and the first partial derivatives guarantee f is differentiable on the circle. However, f is nowhere analytic.

39. $\text{Ln}(1 + i)(1 - i) = \text{Ln}(2) = \log_e 2$; $\text{Ln}(1 + i) = \log_e \sqrt{2} + \frac{\pi}{4}i$; $\text{Ln}(1 - i) = \log_e \sqrt{2} - \frac{\pi}{4}i$.

Therefore,

$$\text{Ln}(1 + i) + \text{Ln}(1 - i) = 2 \log_e \sqrt{2} = \log_e 2 = \text{Ln}(1 + i)(1 - i).$$

40. $\text{Ln} \frac{1 + i}{1 - i} = \text{Ln } i = \log_e 1 + \frac{\pi}{2}i = \frac{\pi}{2}i$; $\text{Ln}(1 + i) = \log_e \sqrt{2} + \frac{\pi}{4}i$; $\text{Ln}(1 - i) = \log_e \sqrt{2} - \frac{\pi}{4}i$.

Therefore,

$$\text{Ln}(1 + i) - \text{Ln}(1 - i) = \frac{\pi}{4}i - \left(-\frac{\pi}{4}i\right) = \frac{\pi}{2}i = \text{Ln} \frac{1 + i}{1 - i}.$$